CorpSum - yet another corpus query and visualization UI

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NoSkE, which is an open-source version of the commercial sketch Engine (SkE) by Lexical Computing Ltd., is a highly efficient and widely used corpus search software providing a wide array of linguistic corpus analysis tools. It is equipped with both a well-defined and -documented REST-API as well as a comprehensive frontend web application named Crystal. Because the Crystal web interface's aim is to provide access to the full bandwidth of functionalities the NoSke query engine is capable of, it can be difficult to navigate for novices. Moreover, some key features such as the visualization of query results and guided colocation and word combination queries have been reserved to its commercial sibling. The latter shortcomings compelled the Austrian Centre for Digital Humanities at the Austrian Academy of Sciences (ACDH-CH) to develop its own bespoke query tool - CorpSum. It is a web application that enables user-friendly, dynamically generated queries in text corpora along different extralinguistic dimensions of variation such as the dimensions time and space. Originally developed to facilitate work with the Austrian Media Corpus (AMC) based on the use of the NoSke Crystal interface by domain experts, the following common analysis tasks were identified and implemented:

- Analysis of Regional Linguistic Varieties - display frequency results divided by regions on geographic map
- Diachronic Lexicographic Analysis - display temporal trends of frequency results and the keyword-in-context in different time periods
- Display sources / topics of frequency results - search and filter by other words using the keyword-in-context results as well as the attached corpus and document metadata

The software is being realized as an open source project using well established technologies and frameworks:

- TypeScript - a strongly typed superset of JavaScript
- VueJS - a performant and versatile framework for building web user interfaces
- D3JS - a JavaScript library for visualizing data

Whilst the initial implementation is bespoke to the use cases of the Austrian Media Corpus, initial tests with other NoSke-hosted corpora have shown that due to it's modular architecture it is already usable on other corpora as well. Because it builds directly on top of the NoSkes swagger-documented REST-API, it can be used on any NoSke instance. In a Poster and Software Demo Session at the DH2025 we hope to get further feedback on our current implementation and gather what kind of documentation domain scientists would require to extend the set of pre-defined analysis tasks themselves.

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